

### Problem-Solution Fit

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| <b>Date</b>          | 27 June 2026                                                 |
| <b>Team ID</b>       | SWTID-2026-5245                                              |
| <b>Project Name</b>  | OptiCrop – Smart Agricultural Production Optimization Engine |
| <b>Maximum Marks</b> | 5 Marks                                                      |

#### Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

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| <p><b>1. CUSTOMER SEGMENT(S) [CS]</b></p> <p>Small and marginal farmers deciding what crop to sow each season.</p> <p>Secondary segment: agricultural researchers and policymakers analyzing crop-environment data.</p>                                                | <p><b>6. CUSTOMER LIMITATIONS</b></p> <p>EG. BUDGET, DEVICES [CL]</p> <p>Limited smartphone/internet access in some rural areas; low digital literacy.</p> <p>Little to no budget for private agronomist consultations; limited access to accurate, timely soil testing.</p>          | <p><b>5. AVAILABLE SOLUTIONS</b></p> <p>PROS &amp; CONS [AS]</p> <p>Local agri-extension officers: personalized but limited availability and reach.</p> <p>Traditional/generational knowledge: trusted but not adaptive to changing soil/climate.</p> <p>Generic govt. advisory apps: broad reach but not personalized to exact soil/weather readings.</p> |
| <p><b>2. PROBLEMS / PAINS + ITS FREQUENCY [PR]</b></p> <p>Farmers struggle to choose the right crop for their soil and climate, often relying on guesswork.</p> <p>Occurs every sowing season (1–3 times/year) and again whenever conditions or crop plans change.</p> | <p><b>9. PROBLEM ROOT / CAUSE [RC]</b></p> <p>No accessible, personalized, data-driven tool combines soil nutrients (N, P, K, pH) with real-time weather to recommend the best crop.</p> <p>Most farmers fall back on tradition or generic advice instead of their field's actual</p> | <p><b>7. BEHAVIOR + ITS INTENSITY [BE]</b></p> <p>Farmers actively get soil tested before sowing (moderate–high intensity, 1–2 times per season).</p> <p>They also consult neighbors and agri officers before finalizing a crop choice.</p>                                                                                                                |

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| <p><b>3. TRIGGERS TO ACT [TR]</b></p> <p>Start of a new sowing season.</p> <p>A poor previous harvest or low yield. Access to a smartphone/soil test report, or advice from an agri officer to try a data-driven tool.</p>        | <p><b>10. YOUR SOLUTION [SL]</b></p> <p>OptiCrop's ML-based recommendation engine takes N, P, K, pH, temperature, humidity and rainfall and instantly recommends the best-suited crop.</p>          | <p><b>8. CHANNELS OF BEHAVIOR [CH]</b></p> <p>ONLINE</p> <p>WhatsApp farming groups, YouTube farming videos, government agri portals/websites.</p> |
| <p><b>4. EMOTIONS</b></p> <p><b>BEFORE / AFTER [EM]</b></p> <p>Before: confused, anxious, financially cautious about choosing the wrong crop.</p> <p>After: relieved, more confident, hopeful about achieving a better yield.</p> | <p>It also provides resource-efficiency tips, directly addressing the root cause by replacing guesswork with a personalized, data-driven prediction — accessible without needing an agronomist.</p> | <p>OFFLINE</p> <p>Local agri-extension offices, seed shops, fellow farmers and cooperative societies</p>                                           |